



8883 - Overlooked Discoveries: An Archival Exploration of Dusty Supernovae from Cycles 2 & 3

Cycle: 4, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

Late-time infrared (IR) observations of supernovae (SNe) with JWST have garnered significant interest in the astronomical community. Spitzer showed that many SN subclasses show IR plateaus lasting for years, or even decades, but the origin and heating mechanism of the dust remained largely unconstrained. Finding the source of the IR plateaus could unlock insights into the Universal dust budget and mass-loss in evolved stars. Only JWST offers the necessary sensitivity and wavelength coverage to characterize the expected cool dust reservoirs in a diverse sample of SNe, particularly at late-times when the IR emission is quite faint. Two recent JWST-enabled papers on SNe 2004et and 2005ip have already revealed surprisingly large cold dust reservoirs, reigniting efforts to investigate dust formation trends in SNe. Several recent JWST proposals target specific SNe with such science in mind, but a larger sample is necessary to explore trends in both time and SN subclass. While dedicated surveys could help build a uniform sample, hundreds of serendipitous observations of nearby SNe (within 200 Mpc) are already archived and set to be made public at the start of Cycle 4, representing low-hanging fruit for this research. Our preliminary work shows that bright SNe have been overlooked, and they are ready to be analyzed. Here, we propose an Archival Research (AR) program to explore these archival observations for surviving dusty SNe or

measure upper limits. Although the dataset may not be uniform, it offers a relatively low-risk and cost-effective approach to enhancing the JWST SNe dust sample and lays a crucial foundation for future monitoring and follow-up studies.

OBSERVING DESCRIPTION

An AR proposal to search archival data at the positions of roughly 300 SNe for old, dusty SNe.